

1 Abstract Interpretation II: Lattices

1. Are the following posets a lattice? A complete lattice?

- $(\mathbb{N}, \leq)$
 - $\inf(a, b) = \min(a, b)$
 - $\sup(a, b) = \max(a, b)$

Are they total functions ?

- Yes for $\min(a, b) = 0$
- No for $\max(a, b) \rightarrow \infty$

So this is a semi-lattice !

But $(\mathbb{N} \cup \{+\infty\}, \leq)$ is a lattice !

- $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$
 - $\inf(a, b) = a \cap b$
 - $\inf_{\forall i}(a_i) = \bigcap a_i$
 - $\sup(a, b) = a \cup b$
 - $\sup_{\forall i}(a_i) = \bigcup a_i$

So this is a complete lattice !

- $(\mathcal{P}(\mathbb{Z}), \subseteq)$
 - $\inf(a, b) = a \cap b$
 - $\inf_{\forall i}(a_i) = \bigcap a_i$
 - $\sup(a, b) = a \cup b$
 - $\sup_{\forall i}(a_i) = \bigcup a_i$

But it is infinite. So this is only a lattice !

- $(E, \leq)$ where E is finite
 - $\inf(a, b) = \min(a, b)$
 - $\sup(a, b) = \max(a, b)$

They are total functions because E is finite:

- $\inf(E) = \min(E)$
- $\sup(E) = \max(E)$